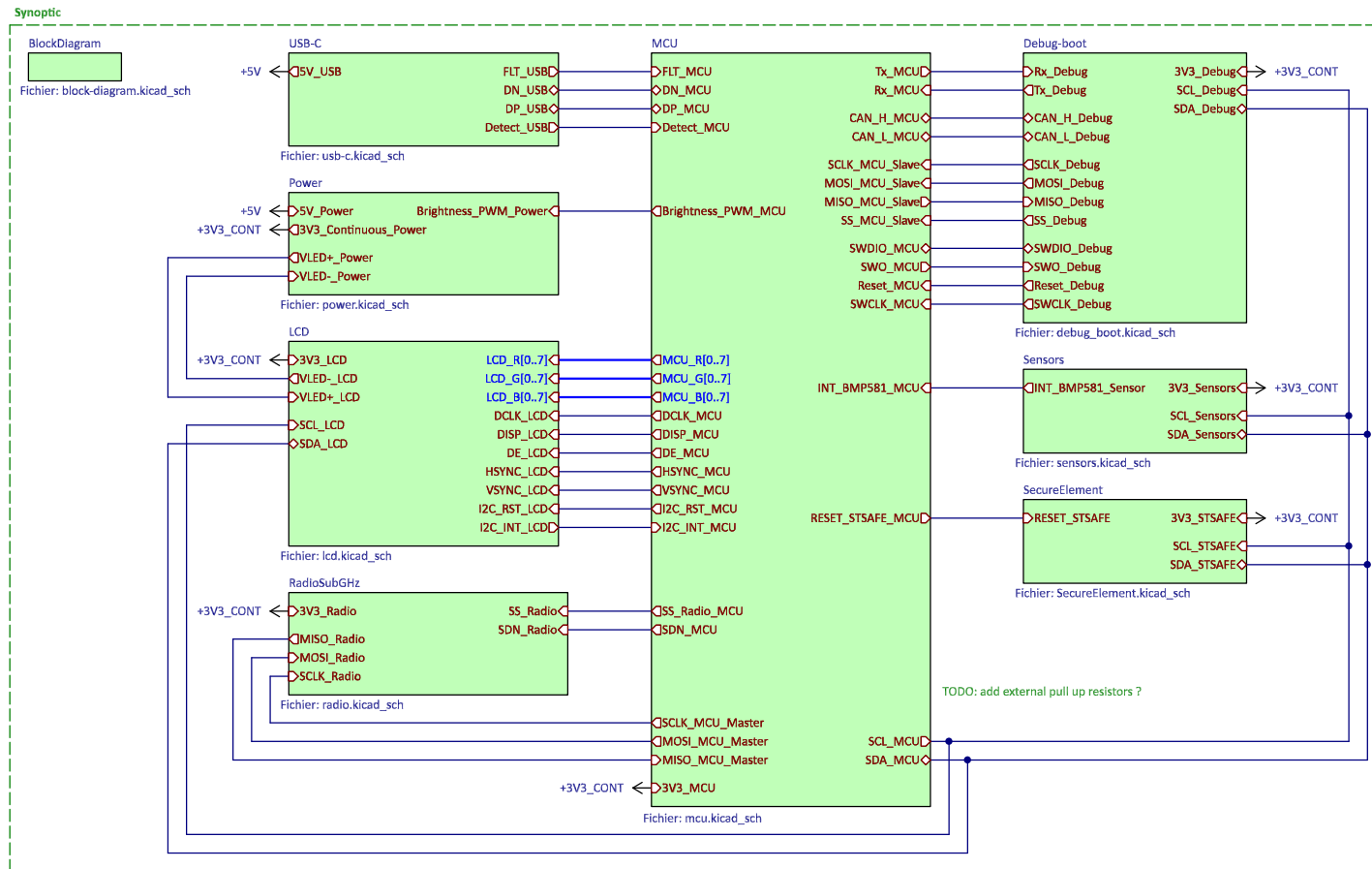


Weather station POC



Revision			
Rev.	Description	Date	Approved
A	Project creation	01/11/2025	JCR

Julien Cruvieux

Sheet: /

File: weather-station-poc.kicad_sch

Title: Weather station POC

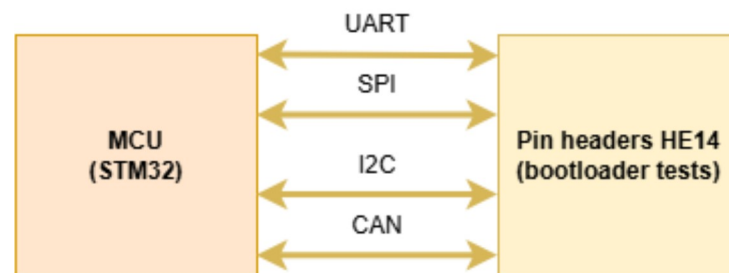
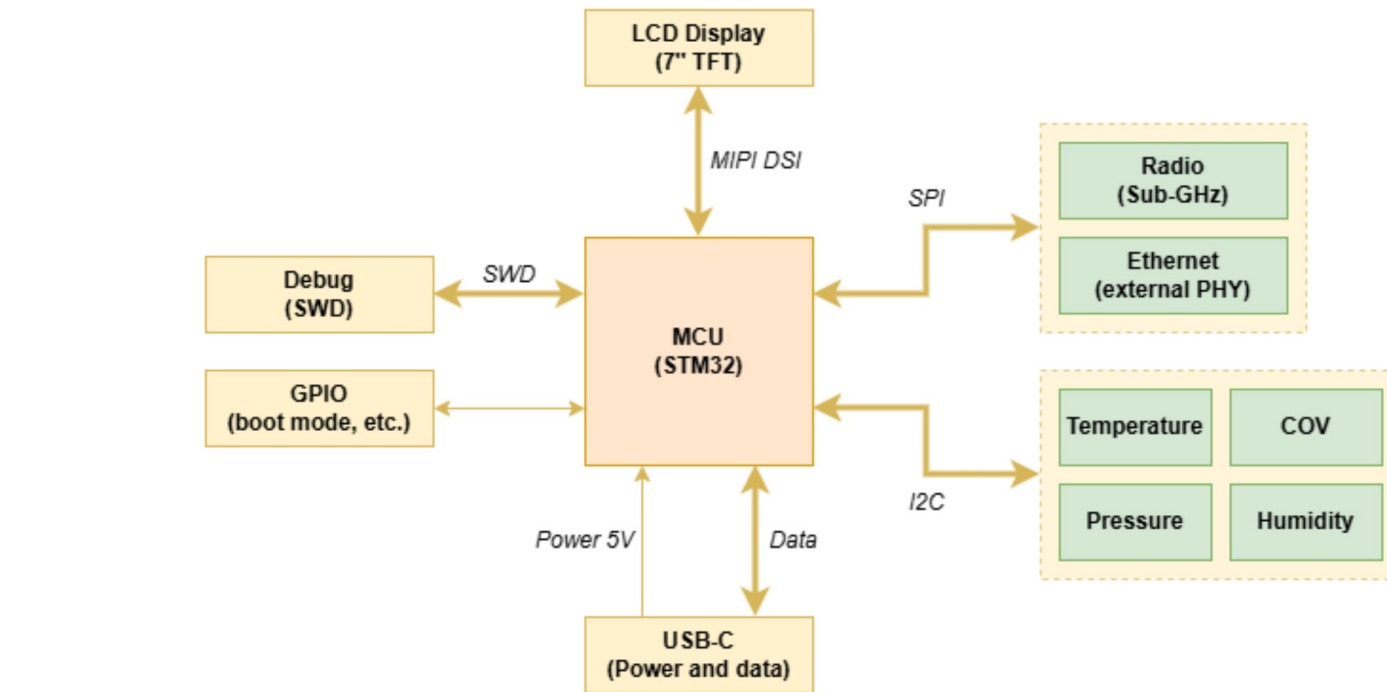
Size: A4

Date: 2025-10-26

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Rev: A

Id: 1/10



Sheet: /BlockDiagram/
File: block-diagram.kicad_sch

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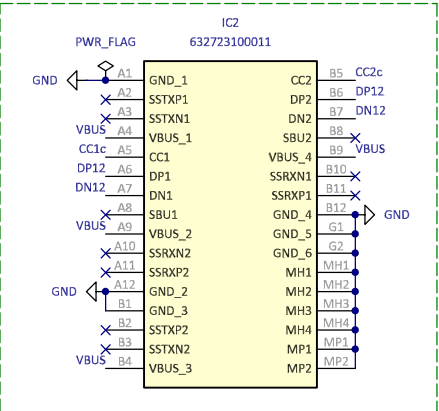
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KiCad E.D.A. 9.0.6

Date:

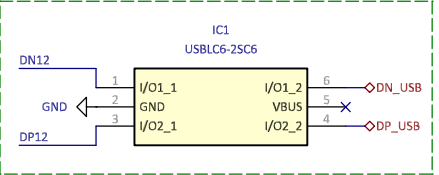
Rev:
Id: 2/10

USB-C

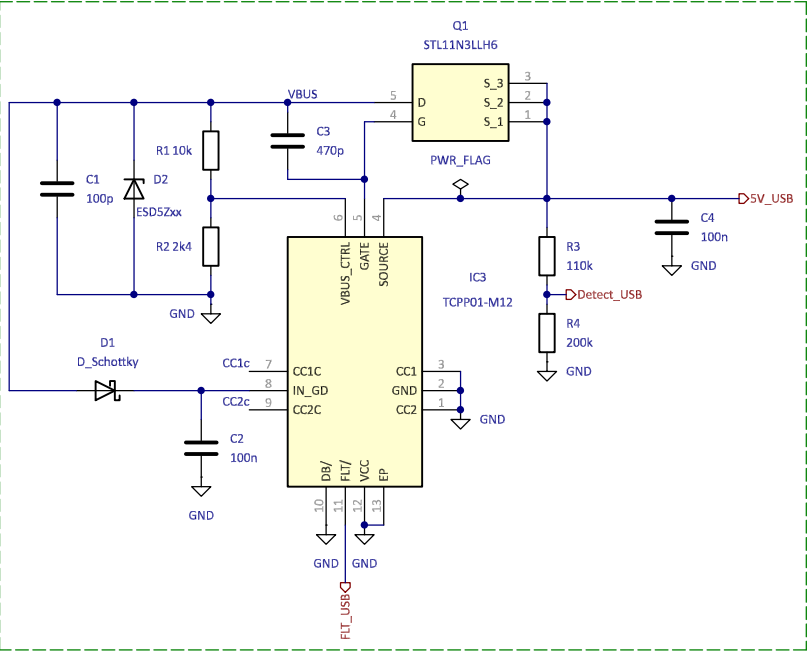
USB-C connector right angle



ESD protection on USB-C data lines



USB-C power sink, 5V, 15W



Sheet: /USB-C/
File: usb-c.kicad_sch

Title:

Size: A4

Date: 2025-11-01

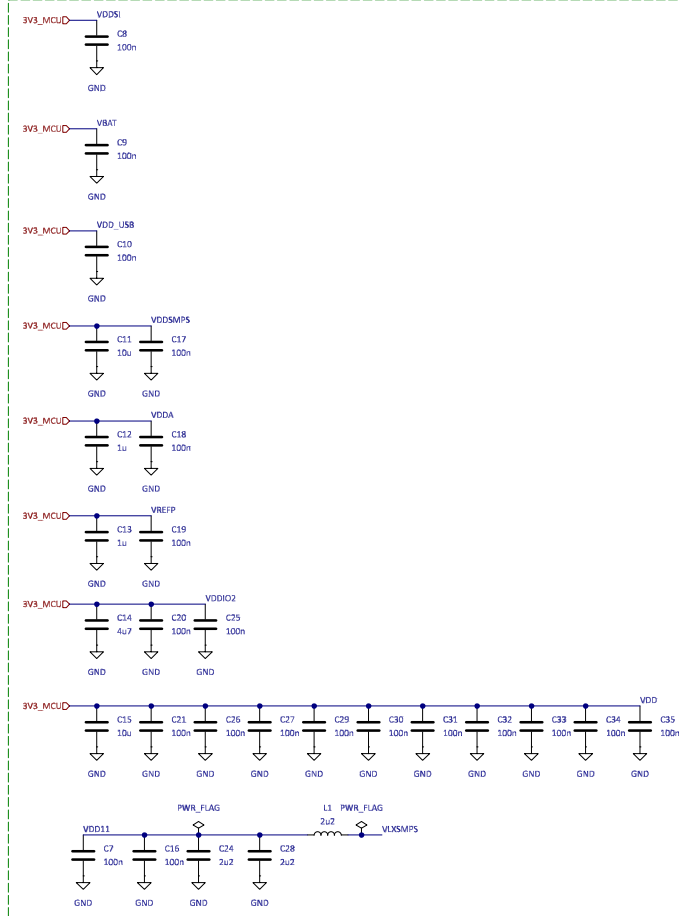
Rev:

KiCad E.D.A. 9.0.6

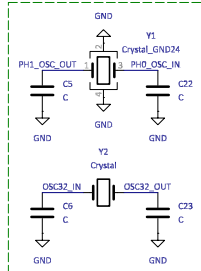
Id: 3/10

MCU

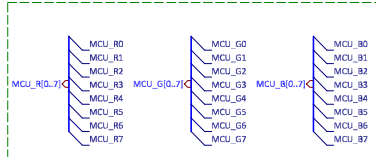
STM32USG821xQ decoupling capacitors



Crystal oscillators (HSE and LSE)

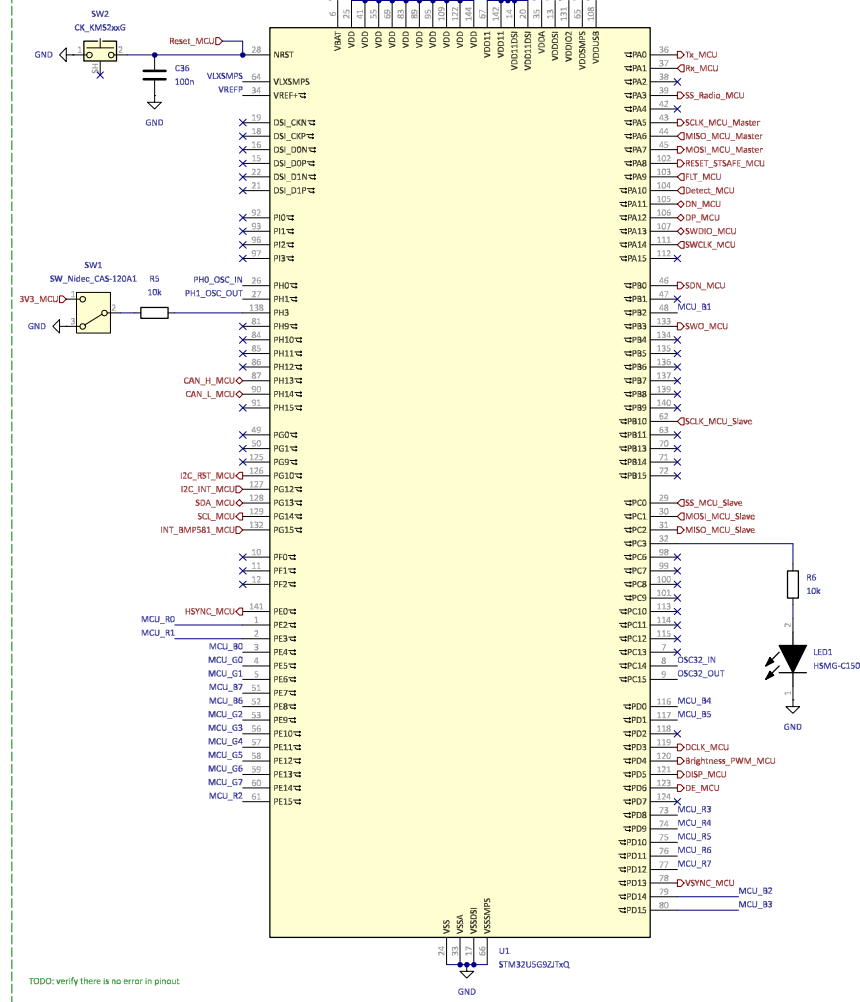


RGB parallel bus



STM32USG821xQ MCU

TODD: add debounce for button



TODD: verify there is no error in pinout

Sheet: /MCU/
File: mcu.kicad_sch

Title:

Size: A3

Date: 2025-11-01

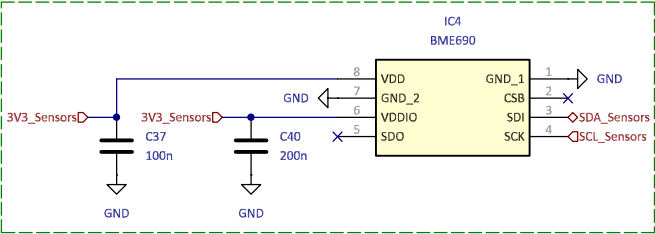
Rev:

KiCad E.D.A. 9.0.6

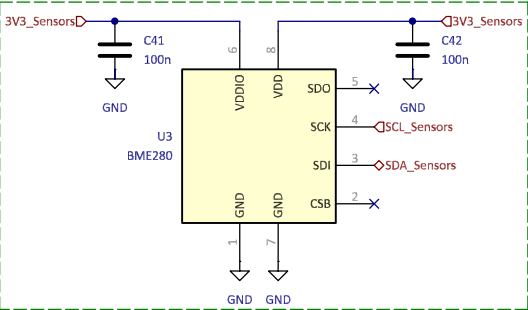
Id: 4/10

Sensors

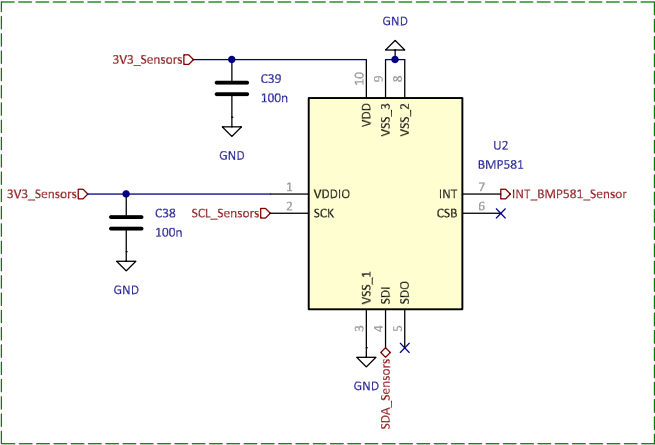
Quality air sensor



Humidity sensor



Pressure and temperature sensor



Sheet: /Sensors/
File: sensors.kicad_sch

Title:

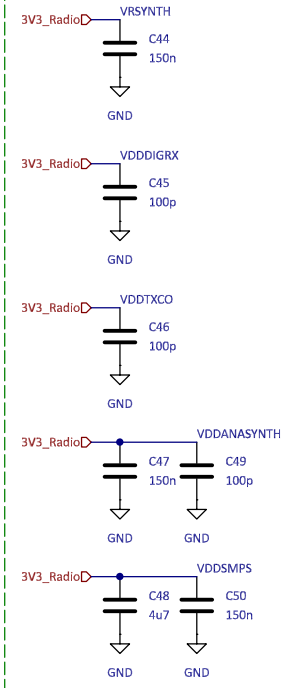
Size: A4
KiCad E.D.A. 9.0.6

Date: 2025-11-01

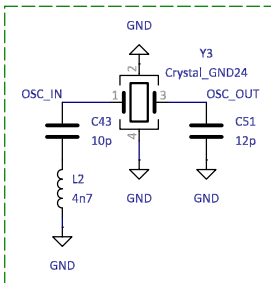
Rev:
Id: 5/10

Sub-GHz radio

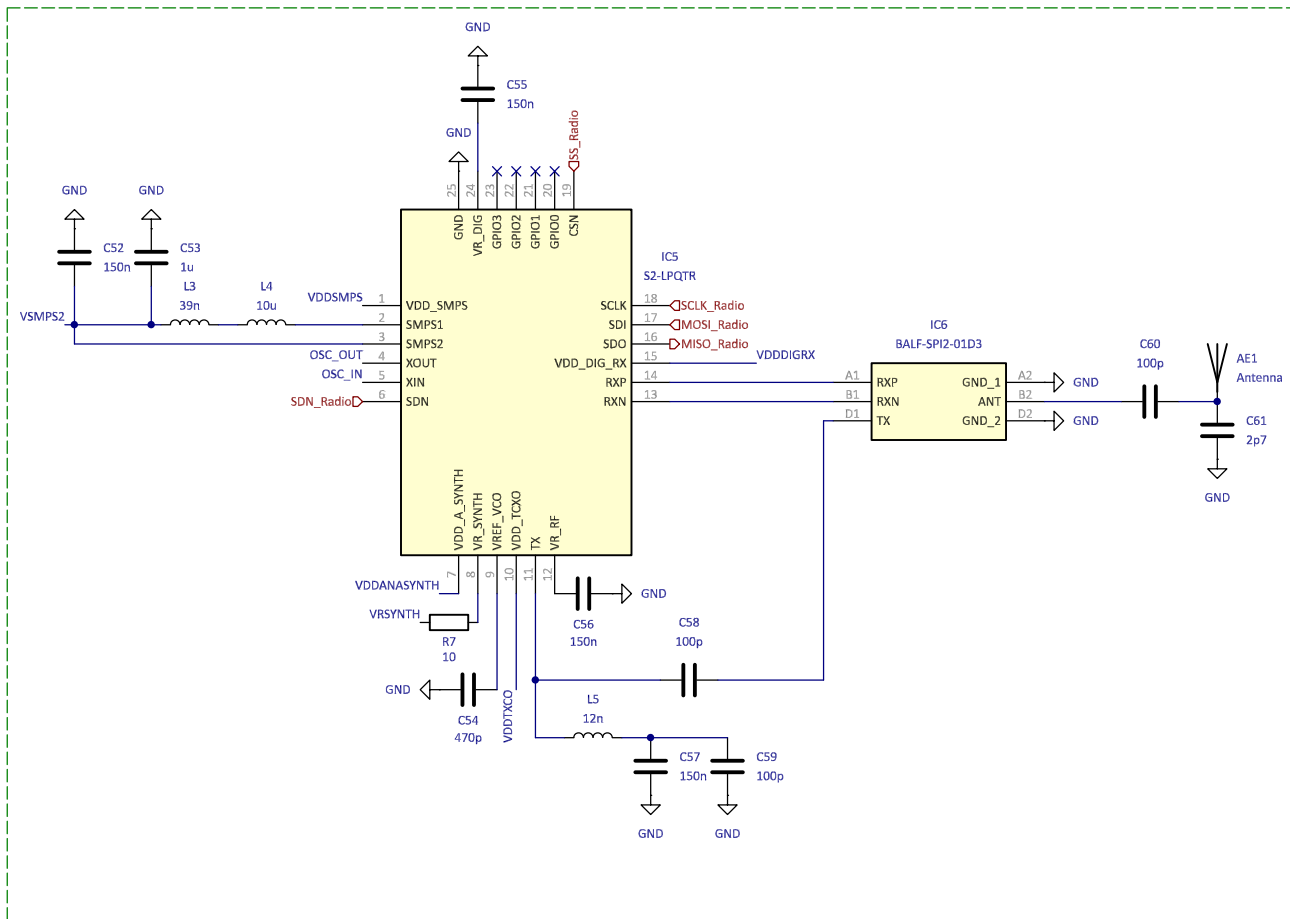
Decoupling capacitors



Crystal oscillator



Sub-GHz module



Sheet: /RadioSubGHz/
File: radio.kicad_sch

Title:

Size: A4

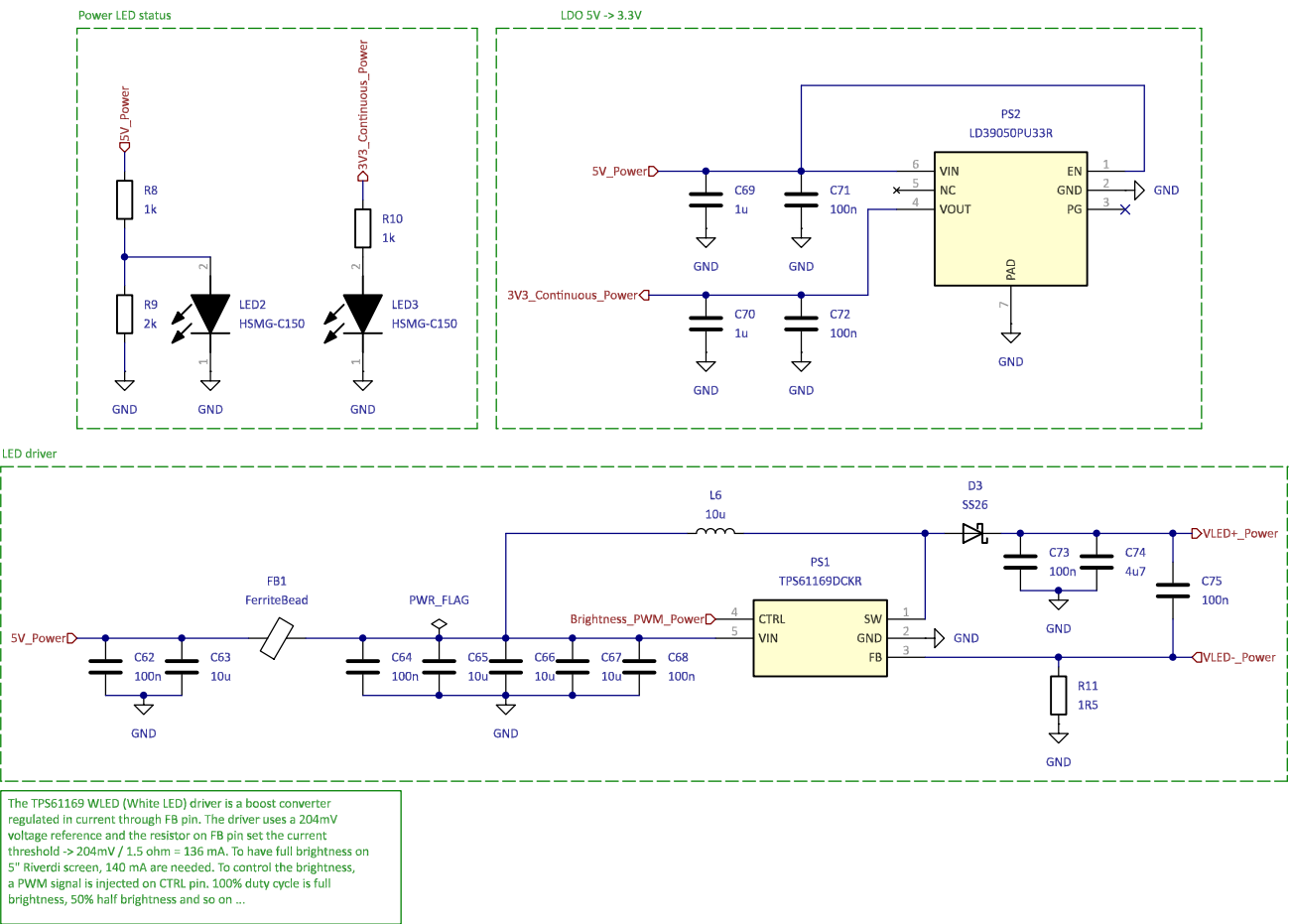
Date: 2025-11-01

KiCad E.D.A. 9.0.6

Rev:

Id: 6/10

Power supplies



Sheet: /Power/
File: power.kicad_sch

Title:

Size: A4

Date:

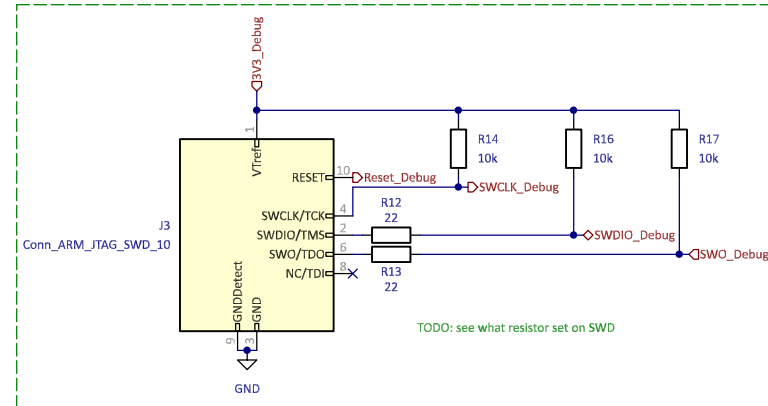
KiCad E.D.A. 9.0.6

Rev:

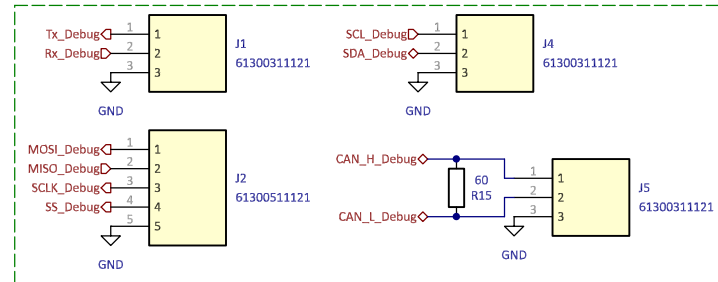
Id: 7/10

Debug

SWD connector (MIP1 10)



Communication interface connectors



Sheet: /Debug-boot/
File: debug_boot.kicad_sch

Title:

Size: A4

Date:

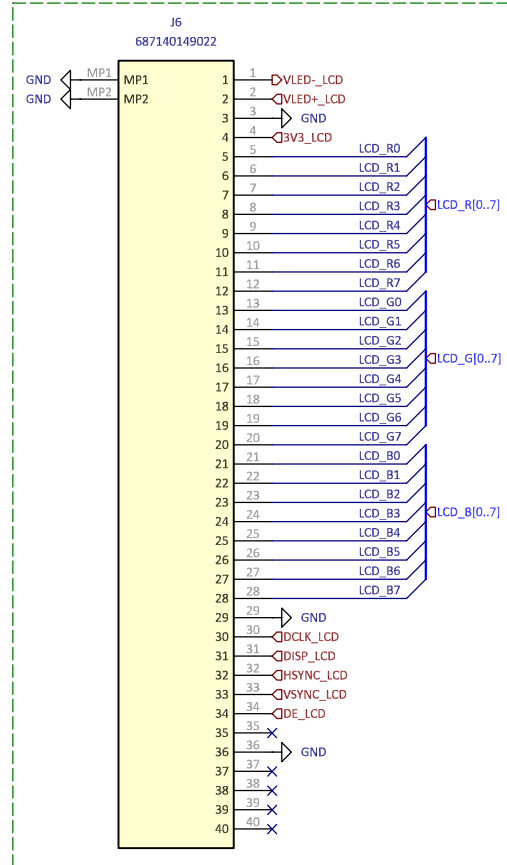
KiCad E.D.A. 9.0.6

Rev:

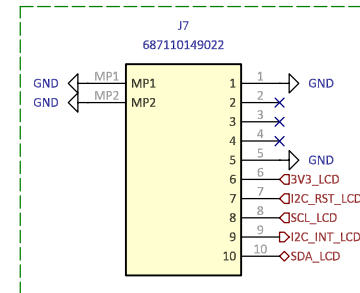
Id: 8/10

TFT LCD screen

TFT LCD 40 pins driver connector



TFT LCD 10 pins touch connector



TODO: think about ferrite beam on LCD connectors

Sheet: /LCD/
File: lcd.kicad_sch

Title:

Size: A4

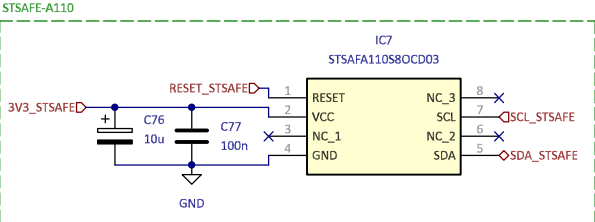
Date:

KiCad E.D.A. 9.0.6

Rev:

Id: 9/10

Secure element (STSAFE)



Sheet: /SecureElement/
File: SecureElement.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 9.0.6

Rev:

Id: 11/10